Essays

|!|BCEEB0310000001|!| ★

With reference to the theory of natural selection, explain how new species of dicotyledonous plants may develop. (11 marks)

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Isolation: (Max. 2)

- A population of a species of dicotyledonous plant is separated into isolated groups by geographic barriers. (1)
- The isolated groups of plants cannot breed with each other. (1)
- The isolated groups of plants may face different environmental conditions. (1)

Natural selection: (Max. 4)

- Plants usually produce huge amounts of pollen / seeds. (1)
- Over-production of offspring intensifies the struggle for existence / competition for the limited resources (e.g. light, water, minerals, etc.). (1)
- There is variation among individuals of the plant species. (1)
- Plants with favourable variations are selected by the environment and have a greater chance to survive and reproduce. (1)
- The favourable variations are passed on to the offspring. Over many generations, the proportion of individuals having the favourable variation increases. (1)

Speciation: (Max. 2)

- Due to natural selection, different characteristics accumulate in each isolated group of plants. (1)
- The isolated group of plants become so different that if they are not isolated again, they cannot interbreed to produce fertile offspring. (1)
- The isolated group of plants have evolved independently and become different plant species. (1)

Communication (Max. 3)

(11 marks)

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Skills assessed	Marks
Understanding basic concepts	8
Communication	3

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